RESEARCH WORKING PAPER / STP-007

A Reconnection Is Not a Recovery

Epoch-Fenced Reacquisition and Recovery Budgets for Unattended Industrial Sensors

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Planned publication: 2026-11-11 **Version:** 1.0 **Status:** WORKING PAPER

DOI: Not assigned (GitHub edition)

Research classification
Architecture and falsifiable evaluation protocol; no empirical performance claims.

> **CENTRAL CLAIM**
> Restoring a communications session is a transport event, not proof that a sensor has resumed trustworthy measurement. A recovery system should fence old acquisition epochs, bound its own corrective actions, and requalify observations before they influence diagnosis.

Abstract

Unattended industrial sensors often fail in ways that invite a simple operational response: reconnect the radio, restart the worker, or power-cycle the node. Such actions can restore a data stream while leaving its epistemic state unresolved. Buffered samples may reappear out of order; a restarted process may inherit stale calibration; a peripheral may advertise while its sensing element remains stuck; and repeated resets can deplete power or conceal a persistent fault. This paper argues that autonomous sensor recovery should be specified as a bounded, evidence-producing protocol rather than a sequence of restart commands. It proposes Epoch-Fenced Reacquisition (EFR): a per-source acquisition epoch that invalidates predecessor streams, a layered recovery ladder with persistent action budgets, and a separate warmup/validation gate before observations regain model authority. The proposal is derived from failure-detector theory, Bluetooth link semantics, operating-system watchdog and restart mechanisms, OT safety guidance, and measurement-uncertainty practice. A fault-injection protocol is specified to compare EFR against naive reconnect and service-level restart baselines. No new recovery algorithm is claimed as validated, and no autonomous firmware or hardware recovery capability is asserted for SynapticSteel TelePod. The contribution is a testable integration contract for remote sensing where physical access may be expensive or impossible.

Keywords: sensor recovery; industrial edge intelligence; epoch fencing; Bluetooth Low Energy; failure detection; watchdogs; recovery budgets; fault containment; measurement validity; unattended telemetry; operational technology.

Suggested citation: Yarian, M. (2026). *A Reconnection Is Not a Recovery: Epoch-Fenced Reacquisition and Recovery Budgets for Unattended Industrial Sensors*. SynapticSteel Technical Papers, STP-007, 1.0. Yarian Works, LLC. GitHub edition; no DOI assigned.

Scope and evidence boundary. This is an architectural synthesis and proposed experiment, not a claim of field-proven reliability improvement. A link-layer timeout, a process restart, and a measurement-quality judgment are distinct events. The proposed policy deliberately keeps them distinct.

1. The operational problem

Consider an accelerometer attached to a pump in an inaccessible shaft. A gateway loses its Bluetooth connection for 40 seconds. The peripheral later reconnects and delivers samples. The operator sees a green icon, yet several questions remain unanswered: Did the sensor reboot? Did its sampling configuration reset? Are the samples live or buffered? Has a previous acquisition worker continued publishing? Did the new session preserve calibration and timestamp origin? A transport indicator cannot answer these questions.

This distinction matters when sensor observations are used to infer machinery condition, rank faults, or trigger a maintenance response. The problem is not simply availability. It is **recovery without false continuity**: preserving the ability to distinguish a new measurement regime from the pre-failure one, even when the device reports plausible numbers immediately after reconnection.

The paper uses *reacquisition* for restoration of a communication or data path, *recovery* for restoration of a bounded measurement service, and *admission* for permission to affect a particular model or claim. These terms are intentionally non-interchangeable. Earlier STP work addresses silence, timestamps, and sensor trust; the present problem concerns what an automated controller may do after failure, and what must be proved before data authority is restored.

2. Prior work and the gap

2.1 Failure detectors report suspicion, not physical truth

Chandra and Toueg formalized failure detectors through completeness and accuracy properties in asynchronous distributed systems [1]. That work is about consensus under crash assumptions, not industrial sensor diagnostics. The transferable lesson is narrower: absence of a timely response does not, by itself, establish the physical cause of nonresponse. A radio outage, host scheduling delay, bus lock, firmware stall, and exhausted battery can produce similar symptoms.

Bluetooth Low Energy specifies a connection supervision timer and rules for declaring an ACL connection lost [2]. The controller can therefore determine that a particular link is no longer viable. It does not certify the transducer, its calibration, or the physical meaning of its output. STMicroelectronics supplies STWIN firmware examples for BLE streaming and USB logging [3], but availability of example transport firmware is not evidence of a production-grade autonomous recovery policy.

2.2 Watchdogs and service managers recover processes, not measurements

Linux watchdog interfaces allow hardware-assisted reset when the expected keepalive fails [4]; Zephyr exposes watchdog setup, timeout installation, and feeding for embedded firmware [5]. The systemd service manager provides process restart policies, watchdog notifications, and start-rate limits [6,7]. These are valuable layers of defense, but each monitors a different predicate: hardware/software liveness, daemon responsiveness, or restart frequency. None independently establishes that an accelerometer has resumed calibrated measurement.

Retry algorithms also have a long engineering history. TCP specifies exponential backoff for retransmission timers [8], while AWS operational guidance explains how retries without backoff and jitter can amplify overload [9]. Applying these ideas to sensor recovery is an engineering adaptation, not a direct guarantee from either source. A retry can worsen a marginal power rail, congest a shared radio, or repeatedly interrupt sensor settling.

2.3 Reliability patterns are necessary but insufficient

Circuit breakers and dependency isolation, exemplified historically by Hystrix [10], bound repeated calls to failing dependencies. NIST OT guidance emphasizes safety, reliability, and performance constraints that ordinary IT recovery designs may overlook [11]. Metrology guidance treats measurement uncertainty as a property requiring evaluation, not something erased by restarting an instrument [12]. Together, these sources motivate a layered contract. The distinctive gap is the *handoff* between successful reconnection and justified re-entry into an evidence pipeline.

3. A fault taxonomy that changes the recovery action

Fault class	Observable symptom	Safer first response
Link interruption	BLE supervision timeout; serial loss	Reacquire link; preserve fault evidence
Worker failure	Process exit, deadlock, stalled progress	Bounded worker restart; new epoch
Peripheral stall	Link alive, no advancing sample sequence	Protocol probe; controlled device reset only if supported
Measurement fault	Fresh but constant/saturated/implausible values	Quarantine measurement; do not infer link fault
Time/provenance fault	Sequence rewind, buffer replay, clock discontinuity	Fence predecessor epoch; re-establish time basis
Power/physical fault	Repeated brownout, missing supply, broken cable	Stop futile retries; degraded state and service request

Table 1. Diagnostic classes are hypotheses, not mutually exclusive ground truth. Escalation must be device-specific and policy-governed.

A heartbeat proves that some software path can respond. It does not prove a sensor is observing its target. Conversely, a silent sensor may be healthy but temporarily unreachable. The recovery policy should preserve multiple hypotheses until discriminating evidence is available.

4. Proposed architecture: Epoch-Fenced Reacquisition

Epoch-Fenced Reacquisition (EFR) is a proposed integration contract, not a novel consensus protocol. Its primary goal is to make an automated recovery action **auditable, bounded, and unable to silently extend the authority of an obsolete stream**. A source-specific supervisor owns a recovery state machine; the data plane carries acquisition identity and validity state to downstream consumers. The supervisor may request permitted corrective actions, but it cannot declare physical health merely because its commands succeeded.

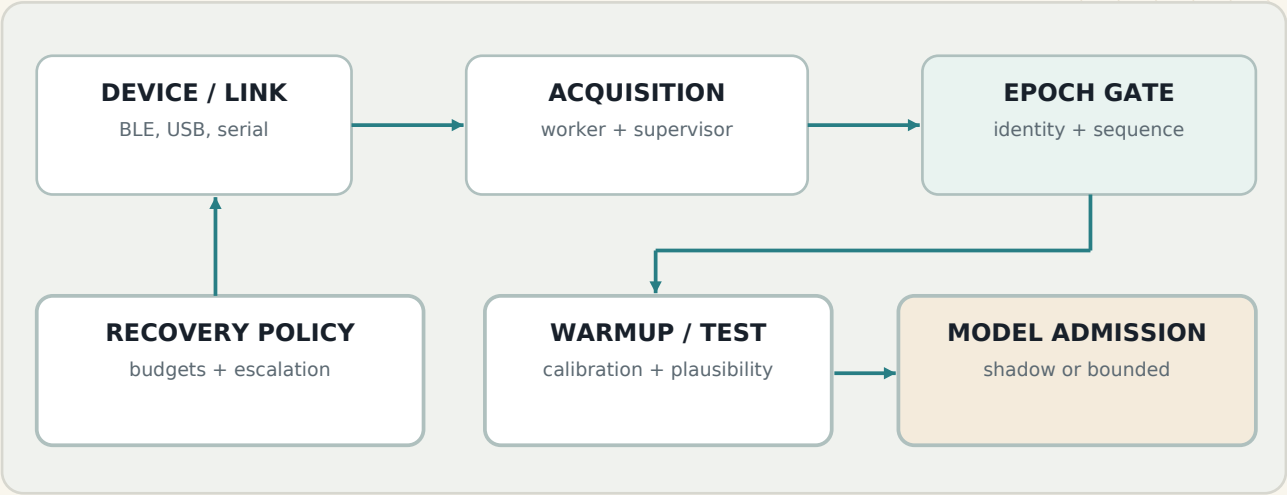


Figure 1. Recovery policy supervises acquisition; the epoch gate feeds measurement validation before model admission. Arrows indicate logical influence, not physical wiring.

4.1 Epoch identity and fencing

Define an acquisition epoch as **E = (device_id, boot_id, session_generation, acquisition_generation)**. Device identity should be authenticated or provisioned where the risk warrants it; a claimed Bluetooth address alone is not sufficient against impersonation. A boot identifier may be a durable monotonic counter or a collision-resistant nonce generated under a documented entropy assumption. Session and acquisition generations advance when ownership changes, including reconnect, worker replacement, or protocol reinitialization. A gateway-issued monotonic fencing token is preferable where multiple workers could race.

Each observation carries **(E, sample_seq, t_observed, t_received, config_digest, quality_flags)**. The admission gate maintains the active epoch and highest accepted sequence for that epoch. Old epochs are retained for evidence but are not eligible for live model input. If the device cannot emit an epoch or sequence, the gateway assigns a local acquisition epoch and explicitly marks the weaker end-to-end guarantee; gateway labeling cannot establish when a buffered sample was physically acquired.

> **SAFETY INVARIANT**
> ADMIT(s, claim) implies active_epoch(s) AND increasing_sequence(s) AND valid_time_support(s) AND permitted_quality_state(s, claim). Reconnection alone never makes ADMIT true.

The gate should commit epoch changes atomically with respect to data admission. A crash between opening a new transport and persisting its generation must fail closed or use a durable, monotonic token. A random session ID can reduce collision risk but does not substitute for correct ownership fencing. An attacker controlling the device can forge its own counters; this architecture is not an authentication or secure-boot substitute.

4.2 Recovery ladder and budgets

The proposed recovery ladder proceeds from least disruptive to most disruptive: (i) observe and classify, (ii) release/reopen the connection, (iii) reinitialize the host protocol stack or worker, (iv) request an explicitly supported peripheral software reset, (v) invoke hardware watchdog or switched power only where the installation provides a safe, authorized mechanism, and (vi) stop and escalate. The ladder is not a promise that every device supports every rung.

For each source, define a rolling-window action budget **B(W) = sum of w(a_i) for actions within W**, where w(a_i) is a configured recovery cost reflecting service interruption, battery/energy use, wear, and risk. Also enforce absolute ceilings for high-impact actions, minimum spacing, and a persistent reboot-loop counter. When the budget is exhausted, enter QUARANTINED rather than repeatedly attempting the same intervention. Weighting is a policy design choice requiring local validation; no universal safe numerical threshold is proposed.

Exponential backoff with randomized jitter reduces synchronized retry storms, but backoff cannot repair a missing power supply or a mechanically failed transducer. A token-bucket budget can be useful for limiting frequency; for safety-critical resets, explicit per-action quotas and escalation are clearer. Restart limits that disappear when a service unit is unloaded or a host reboots must not be treated as a durable field-service budget [7].

4.3 State transitions and admission

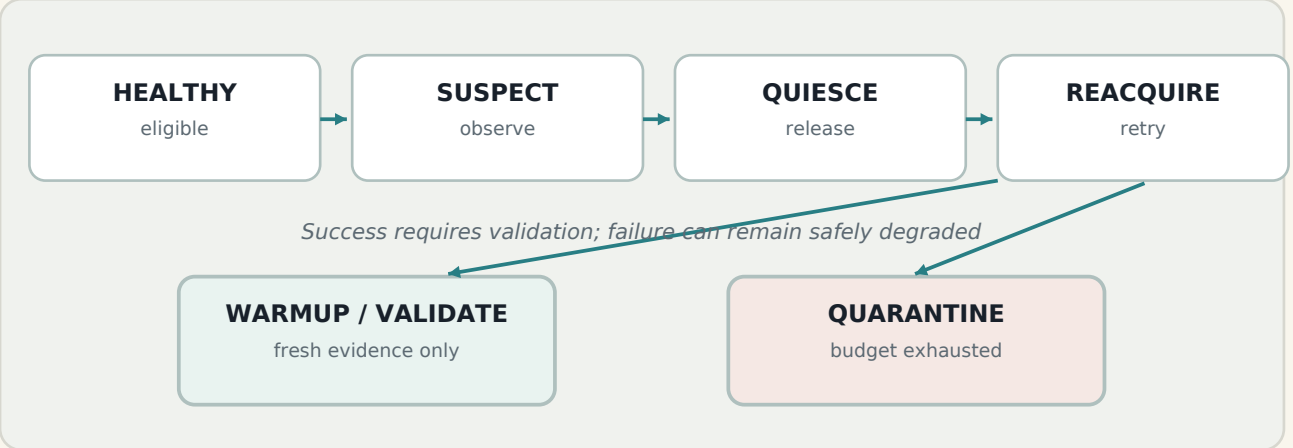


Figure 2. A compact state machine. WARMUP/VALIDATE may return to HEALTHY only after the measurement-specific checks pass; otherwise the source remains degraded or quarantined.

A minimal state model is HEALTHY, SUSPECT, QUIESCING, REACQUIRING, WARMUP, VALIDATING, QUARANTINED, and AWAITING_SERVICE. The state machine records cause codes and every action attempt. On a new epoch, the gate revokes the previous epoch immediately, discards or archives late predecessor packets, and places the new stream in shadow until qualification. Where validation is impossible, a system may expose data as explicitly *unverified* for operator context without granting it diagnostic authority.

Warmup is not a fixed delay by definition. It may require stable configuration, advancing sample counters, sufficient temporal coverage, sensor-specific stabilization, independent plausibility evidence, and an established timestamp origin. Calibration may survive a restart, be invalidated by it, or be unknown; that is a device-specific fact to be represented rather than assumed. GUM uncertainty [12] informs measurement confidence but does not directly specify recovery eligibility.

5. Formal claims, assumptions, and counterexamples

5.1 Safety is achievable under explicit boundaries

The proposed safety claim is local and conditional: if every downstream consumer accepts data only through a correctly implemented epoch gate, and the gate uses durable ownership state, no packet from a superseded acquisition epoch will be admitted as current. This is a software invariant suitable for property-based testing or model checking. It does **not** prove the active sensor measures the true physical quantity.

Counterexample A: a device reconnects and replays ten seconds of buffered samples with a fresh gateway epoch. Epoch fencing alone accepts them unless source acquisition time or buffer status is verified. Counterexample B: two workers each believe they own the sensor because the epoch update is not atomic. A durable fencing token and single admission owner are required. Counterexample C: a sensor reports a monotonically increasing sequence while the transducer is stuck. Temporal validity passes, but measurement validity fails. These are reasons to retain independent admission predicates.

5.2 Liveness is not unconditional

A bounded liveness claim is possible only if the failure is transient, communication and power eventually return, a supported recovery action exists, the sensor can meet validation criteria, and the recovery budget permits further attempts. A deeply buried sensor with a broken cable or destroyed battery cannot be recovered by a host-side restart. Physical reachability and firmware cooperation are assumptions, not software features.

5.3 Recovery can itself be a hazard

Frequent reconnects can increase radio occupancy; peripheral resets can erase diagnostic state; power cycling can create inrush or mechanical side effects; and indiscriminate watchdog feeding can hide a hung acquisition loop. A watchdog should be driven by a meaningful progress predicate rather than merely by a timer thread. In high-consequence OT environments, remote reset and power switching must be assessed against equipment safety constraints and human authority [11].

6. A minimal interoperable evidence contract

The following record is illustrative, not a deployed API. Field names are deliberately small enough for an edge node. A production implementation would version its schema, define clock domains and units, authenticate origins, and document which fields are observed, derived, or policy-assigned.

```
{ "source_id": "vibration-01", "epoch": {"boot_id":"b42", "session":17, "acq":3},
  "sample_seq": 812, "transport": "BLE", "link_state":"connected",
  "measurement_state": "WARMUP", "model_admission": "SHADOW_ONLY",
  "observed_monotonic_ns": 492001120000, "received_monotonic_ns": 492001131000,
  "config_digest": "sha256:...", "calibration_state":"UNVERIFIED",
  "recovery": {"action":"reconnect", "attempt":2, "budget_remaining":4},
```

"reason_codes": ["EPOCH_CHANGED", "VALIDATION_PENDING"] }

The record distinguishes a live connection from a validated measurement and from claim-specific model admission. A real implementation should use typed schemas rather than a textual digest placeholder, and should preserve raw source evidence separately from derived reason codes. If persistent storage is constrained, a bounded append-only journal with rollover policy is preferable to an unbounded log.

7. Competing designs and why a hybrid may be justified

Design	Strength	Blind spot
Unconditional reconnect	Simple; low software overhead	Can loop forever; restores link without qualification
systemd restart limits	Mature process supervision; cheap to deploy	Process-centric; may reset on host reboot; no sample epoch
Device hardware watchdog	Recovers some firmware hangs without gateway	Cannot repair broken power/physics; may erase evidence
Cloud-managed fleet agent	Central policies, audit, remote diagnostics	Depends on connectivity; latency and security boundary
EFR proposal	Epoch safety + bounded actions + quality re-entry	More state, test burden, persistent storage and policy complexity

Table 2. Competing approaches are not mutually exclusive. EFR should reuse the lower-level watchdog and supervisor mechanisms rather than replace them.

A strong counterargument is that a mature service manager, an application-level heartbeat, and good calibration procedures already solve most practical cases. For low-cost, noncritical, locally accessible sensors, that may be correct: EFR may impose unjustified complexity. The proposed architecture is most plausible where sensor visits are costly, reconnection churn is common, or downstream decisions have consequences. Its incremental benefit must be measured against a well-configured conventional baseline, not against an intentionally fragile straw man.

8. Falsifiable evaluation plan

A useful evaluation should isolate the *recovery mechanism* from the *measurement claim*. Use the same sensor, firmware, gateway, and workload across randomized trials, and compare three policies: A, naive reconnect; B, systemd-style bounded worker restart plus standard driver behavior; and C, EFR with epoch fencing, budgeted escalation, and revalidation. Keep the physical system safe and separate software-injected faults from real hardware faults.

8.1 Fault matrix and ground truth

- **Transient link:** RF attenuation or cable interruption, with independent timestamps of outage and restoration; distinguish reconnect latency from valid-data latency.

- **Host acquisition:** worker crash, deadlock, orphaned child, duplicate collector, and stale file descriptor; verify no obsolete epoch is admitted.
- **Peripheral behavior:** controlled firmware reset, advertising without sample progression, and simulated sensor-register misconfiguration; preserve safety and vendor constraints.
- **Data and clock:** delayed buffer replay, sequence rewind, host clock step, timestamp wrap, and stale calibration token; check evidence lineage and shadow admission.
- **Power and correlated failure:** brownout simulation in a lab fixture, shared gateway outage, and simultaneous reconnect storm; measure recovery budgets and aggregate radio load.

For each fault, record independently witnessed ground truth (e.g., power monitor, host process trace, RF capture, reference sensor) and label uncertainty where the witness is imperfect. An emulated BLE link is useful for deterministic tests but cannot stand in for physical RF interference, sensor warmup, or electrical power cycling.

8.2 Primary outcomes and decision thresholds

Pre-register the primary safety outcome as **false-healthy exposure**: total duration in which data are admitted as validated despite known replay, wrong epoch, failed calibration, or stale source state. Secondary outcomes include p50/p95 time to valid data, fraction of recoverable faults resolved without site visit, reset count per day, energy per recovered hour, false quarantine rate, diagnostic evidence retained, CPU/memory overhead, and operator intervention burden.

A defensible go/no-go rule would require zero known stale-epoch admissions in adversarial tests, bounded worst-case reset counts under persistent faults, and a measured improvement in false-healthy exposure without an unacceptable increase in time to valid data or energy consumption. Numerical acceptance thresholds should be set from device and application requirements before running trials; none are asserted here. Confidence intervals should be reported per fault class and device, with repeated trials and negative results included.

8.3 Ablation and sensitivity

Remove epoch fencing, recovery budgets, and measurement revalidation one at a time. This determines whether each mechanism contributes measurable value or merely increases complexity. Vary BLE supervision timeout, backoff caps, energy budget, and validation window. Repeat across at least two sensor families and one wired interface before claiming transport-general behavior. Test both benign failures and compromised or malformed telemetry separately; fault tolerance is not equivalent to adversarial security.

9. Experimental scope and implementation boundary

This paper is a research proposal and evaluation protocol. Private operational checkpoints are excluded from this public edition. No implementation, deployment, field validation, or measured performance benefit is established by the proposal. A prototype should first be evaluated with recorded or simulated inputs, explicit baselines, failure cases, and independently measured outcomes.

10. Limitations and research implications

EFR cannot identify every failure mode. It may not detect a plausible stuck-at value, a slowly drifting transducer, or a compromised peripheral that emits fresh-looking sequences. Epoch counters require durable semantics, secure identity where appropriate, and atomic handoff; otherwise fencing can be bypassed by implementation error. Recovery budgets are policy parameters whose optimal values vary with battery chemistry, radio environment, device wear, and process criticality. Warmup validation can suppress genuine anomalies immediately after a restart, so a separate non-authoritative evidence lane is necessary to avoid silently discarding novel physical events.

The architecture also creates a new software component that can fail. A supervisor with a corrupt state journal, bad clock assumptions, or a broken escalation rule may reduce availability relative to a simpler service manager. The proposed remedy is not complexity for its own sake, but a small explicit contract whose invariants can be tested, monitored, and rolled back. Independent replication and field data are required before any quantitative reliability or cost-of-service claims.

The research contribution is strongest as a protocol-and-evaluation paper. An SSRN working-paper release may be suitable after editorial review. An arXiv submission, conference paper, or journal article would require reproducible implementation details, registered or clearly described experimental design, multi-device results, uncertainty analysis, and comparison with competent baselines. No venue acceptance or peer review is implied.

11. Conclusion

The simplest recovery dashboard asks whether a sensor is connected. A dependable industrial evidence system must ask a harder question: *what has actually been re-established, under which acquisition identity, after which interventions, and with what remaining uncertainty?* The proposed Epoch-Fenced Reacquisition contract separates link restoration from measurement qualification, limits automated corrective action, and preserves discontinuities for downstream diagnosis. Its value remains a hypothesis until controlled tests show that it reduces false continuity without imposing excessive latency, energy use, or operational complexity.

Related SynapticSteel Technical Papers

STP-001, Failure Metabolism. Historical operational memory and evidence-preserving intervention context; a Zenodo identifier has been reported in project records but is not independently asserted here as a verified public release.

STP-003, The Timestamp Is Not the Event. Temporal provenance and uncertainty; EFR makes the acquisition boundary explicit after restart.

STP-004, Silence Is a Signal, But Not a Measurement. Missingness as evidence about observability; EFR concerns bounded corrective action and subsequent readmission.

STP-005, Trust Is Not Agreement. Claim-specific sensor credibility and admission; EFR supplies a post-recovery identity and qualification boundary.

STP-006, The Probe Is Part of the Process. Observation-induced perturbation; resets and repeated reconnection attempts are also interventions that may alter the monitored system.

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Source assessment. References [1] and [8] are peer-reviewed research/standards-track work; [2], [4]-[7], [11], and [12] are primary technical specifications, documentation, or guidance; [3], [9], and [10] are vendor/project sources used for implementation context, not independent proof of EFR effectiveness. All URLs were checked during preparation. No experimental results are asserted.

Publication and provenance note

Author-released working paper. GitHub edition published 11 November 2026. Not peer reviewed; no empirical performance improvement or production deployment is established. No DOI is assigned to this GitHub edition. Cite its versioned repository record. Copyright Mark Yarian. All rights reserved; public reading and citation permitted.